

Laser-Induced Forward Transfer of Photonic Integrated Circuits

N. de Louw⁽¹⁾, G. Arutinov⁽¹⁾, A. Abdou⁽¹⁾, G. Gelinck⁽¹⁾, M. Giesbers⁽¹⁾, S. Roso Casares⁽¹⁾, A. Karki⁽¹⁾, H. Fledderus⁽¹⁾, R. Ollearo⁽¹⁾

⁽¹⁾ TNO/PITC, High Tech Campus 31, Eindhoven, The Netherlands, nick.delouw@tno.nl

Abstract We present laser-induced forward transfer (LIFT) as a novel die-to-wafer assembly approach for photonic integrated circuits (PICs). Placement accuracies approaching 500 nm (3σ) are achieved, alongside efficient electro-optical coupling, underscoring LIFT's promise for high-throughput, high-precision PIC integration. ©2026 The Author(s)

Introduction

Driven by the recent advancements and widespread deployment of artificial intelligence (AI) technologies, the demands of data-intensive systems have increased to a level that exceeds the performance capabilities of any single photonic or electronic chip platform. No single material platform is capable of simultaneously delivering the bandwidth, efficiency, functionality and scalability required for these emerging applications. Consequently, heterogeneous and hybrid integration strategies are emerging, in which multiple specialized chip platforms are co-integrated within a single package to exploit their respective strengths [1,2]. This approach enables the realization of high-performance, multifunctional architectures that cannot be achieved through monolithic integration alone.

However, as these system requirements evolve, the limitations of state-of-the-art assembly and integration technologies are becoming the critical bottleneck. Chip assembly is still largely based on mechanical pick-and-place. This contact-based approach works well for robust and mechanically tolerant components. However, when applied to next-generation chips, which are becoming smaller but also thinner, pick-and-place technology faces fundamental limitations. Mechanical handling increases the risk of damaging such ultra-thin dies. Moreover, the throughput remains limited, even in the most advanced automated machines, because the process is inherently sequential. As a result,

these methods struggle to scale to the integration densities required for next-generation chip and system architectures, underscoring the need for alternative chip assembly methods that support greater parallelization, such as micro-transfer printing (MTP) [3].

In this paper, we present a novel chip assembly method for high-throughput, high-precision transfer of photonic integrated circuits (PICs). While the underlying assembly method, laser-induced forward transfer (LIFT), has been developed for other application markets, such as solder paste printing (Keiron Technologies [4]) and microLED transfer [5,6], we present its first implementation for the integration of PICs, demonstrating efficient electro-optical coupling of an indium phosphide (InP) photodiode (PD) to a silicon-on-insulator (SOI) PIC. One key challenge lies in achieving the stringent placement accuracy (< 1 micrometer) needed to effectively couple light from one PIC to another.

Laser-Induced Forward Transfer (LIFT)

Figure 1 presents the overall LIFT process flow. Initially, all PIC coupons are simultaneously transferred from the native source wafer onto the glass carrier with the TNO proprietary release stack. This process, referred to as the pick-up step, currently employs suspended PIC coupon structures similar to those used for MTP [7]. The populated glass carrier is then aligned with the target substrate, maintaining a typical spacing of several tens of micrometers between the chips

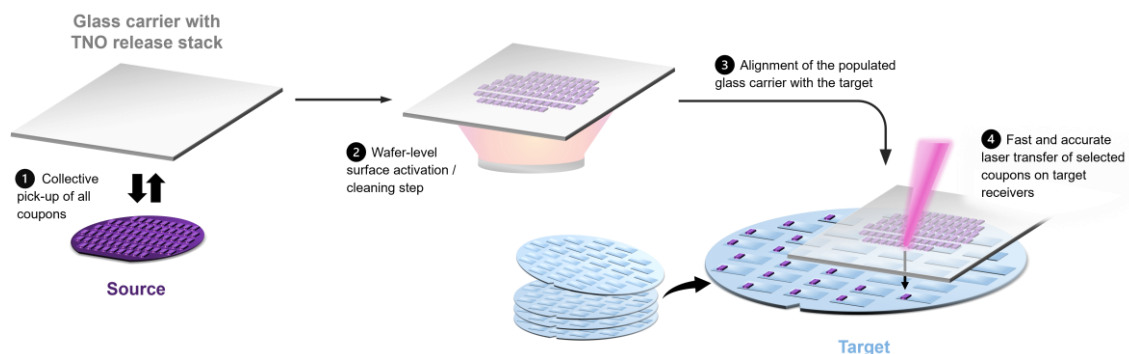


Fig. 1: LIFT process flow for the transfer of (PIC) coupons: collective pick-up (1), optional wafer-level surface activation or cleaning (2), alignment (3), laser-assisted transfer on (multiple) target(s) (4).

and the target substrate. Following alignment, a laser pulse is focused on the backside of the chip, specifically the InP coupon attached to the TNO release stack. This pulse generates a driving force that releases the chip from the carrier, enabling its transfer onto the underlying target substrates. The combination of the release stack carrier buildup and optical design ensures that the release stack absorbs the vast majority of the incident laser pulse energy, thereby maintaining the thermomechanical stability of fragile components during release.

The laser-assisted release mechanism operates at μs timescales per die, enabling extremely high intrinsic throughputs in the order of 10s of million units per hour. In addition, the selective release process enables a known good die (KGD) approach and offers flexibility in pitch conversion from source wafer to target substrate, which can translate into efficient wafer usage.

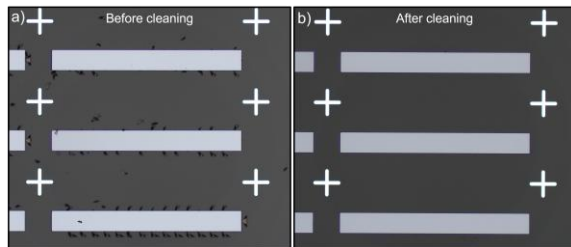


Fig. 2: Glass carrier with TNO release stack with picked-up InP coupons ($80 \times 800 \mu\text{m}^2$) before (a) and after (b) wafer-level cleaning step.

Another unique aspect of our contact-less mass transfer technology, compared to other state-of-the-art methods, is its compatibility with an intermediate surface treatment step, which can be applied to the components at a wafer level between the pick-up and release steps, for cleaning and activation purposes. This distinctive feature is essential for enabling adhesive-less, direct bonding, as it facilitates the removal of debris and particles accumulated during previous processing steps prior to the transfer, such as

contamination introduced during coupon fabrication or debris generated during pick-up. This effect is clearly depicted in Figure 2, which shows picked-up InP coupons ($80 \times 800 \mu\text{m}^2$) onto the glass carrier with release stack, before and after an additional wafer-level cleaning, resulting in clean interfaces required for high quality bonding.

Results

Figure 3 shows examples of (selectively) transferred mechanical InP coupons with a wide range of dimensions and aspect ratios. No indications of cracking, damage or breakage were detected in the transferred coupons. The deviations observed in some $120 \times 120 \mu\text{m}^2$ coupons were already present prior to transfer. These results highlight the scalability of the LIFT process, as there is no fundamental limitation to extending it to more extreme dimensions and higher aspect ratios. Rather, the dimensions shown in this figure reflect the coupon sizes available at the time of the study.

Following the demonstration of process compatibility, a proprietary method was developed that combines alignment features with an acceptor material system to achieve high accuracy, adhesive-less bonding of InP coupons to functional target samples. Figure 4 presents the results of a statistical study on the placement accuracy of InP PD coupons ($40 \times 50 \mu\text{m}^2$) transferred onto a silicon target sample using LIFT. Displacement values of 480 nm (3σ) and 680 nm (3σ) are achieved in the y- and x-directions, respectively, demonstrating that LIFT can fulfil the demanding alignment accuracy requirements for integrated photonics applications.

Finally, to further validate the LIFT assembly, we integrated an InP PIN PD onto an SOI waveguide, directly bonded without additional adhesive layers, and measured the electro-optical coupling. The InP PD was placed with an accuracy of 485 nm in the direction transverse to the light propagation, which is the most critical

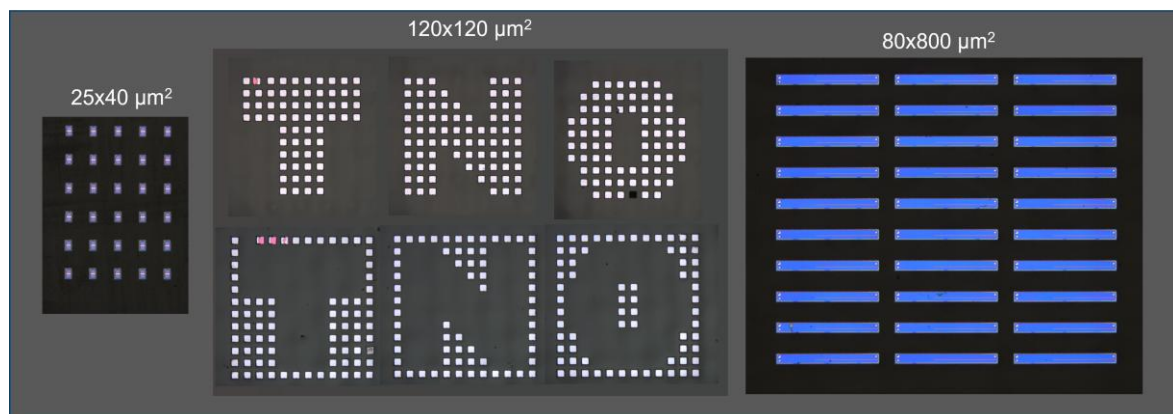


Fig. 3: Transferred mechanical InP coupons using LIFT process flow for a range of dimensions and aspect ratios ($25 \times 40 \mu\text{m}^2$, $120 \times 120 \mu\text{m}^2$ and $80 \times 800 \mu\text{m}^2$), also showing the selectivity aspect in the transferred $120 \times 120 \mu\text{m}^2$ coupons.

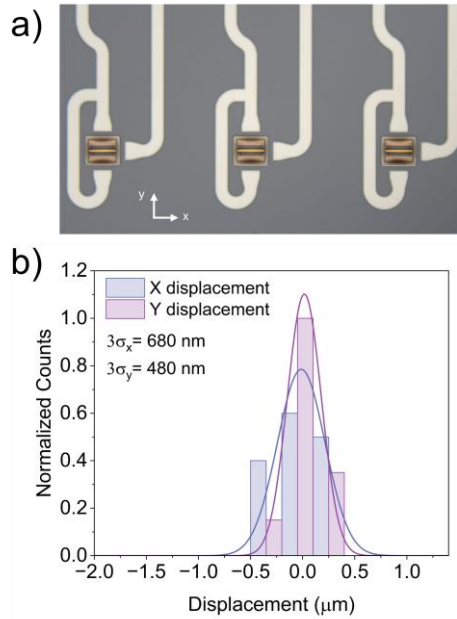


Fig. 4: LIFT placement accuracy of InP PD coupons ($40 \times 50 \mu\text{m}^2$): microscope image (a) and statistical distribution (b).

direction for efficient optical coupling. For this demonstration, the SOI receiver PIC was designed to provide optimal evanescent coupling to the InP PD coupons, which are $40 \times 50 \mu\text{m}^2$ in size with an active area of $5.5 \times 7.5 \mu\text{m}^2$. The standard “MPW 220nm SOI passive” platform from Cornerstone was selected for the SOI PIC [8].

Figure 5 shows the experimental setup that is used to measure the electro-optical coupling between the InP PD and SOI waveguide after LIFT integration. A tunable laser source (1550 nm) is connected, via a polarization controller, into a single SMF28e+ fiber. The light is then coupled from the fiber into a standard grating coupler (from Cornerstone’s PDK [8]) on the SOI PIC. The light that is evanescently coupled from the SOI

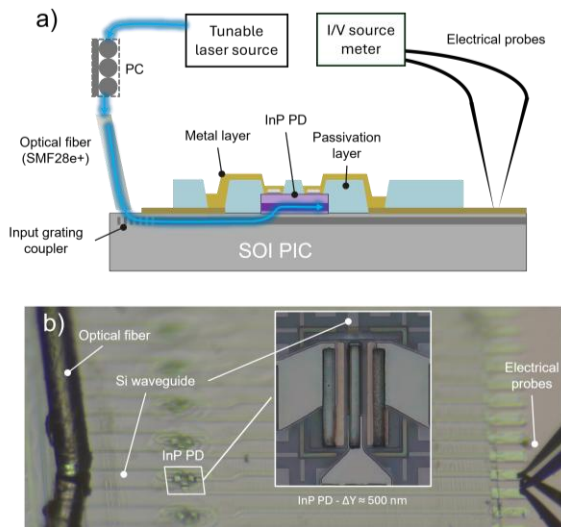


Fig. 5: Experimental setup for electro-optical characterization measurements of the LIFT integrated InP PD onto the SOI PIC: schematic (a) and microscope image (b).

waveguide into the PD is then converted into a photocurrent under reverse bias. After the required post-processing of a passivation and metallization layer, the PD is electrically contacted by probing the bond pads of the SOI PIC.

Figure 6 presents the measured photocurrent of the integrated PD as a function of the optical fiber output power at a reverse bias of -0.6V , thereby demonstrating the electro-optical coupling between the PD and the waveguide. The measurements result in a fiber responsivity of 0.17 A/W , which thus comprises the grating coupler efficiency, the evanescent coupling efficiency and the intrinsic PD responsivity. Assuming a grating coupler efficiency of $5.5 - 6.5 \text{ dB}$ [8], the evanescent coupling efficiency in combination with the intrinsic PD responsivity leads to a measured system responsivity of $0.60 - 0.75 \text{ A/W}$, which is in agreement with the measured responsivity from a previous study of an InP PD with a similar epitaxial layer structure (where the difference is a 150 nm thick n-contact layer, instead of 700 nm) [9]. These results therefore demonstrate efficient electro-optical coupling upon LIFT assembly, highlighting the applicability of LIFT as an integration technique for PICs.

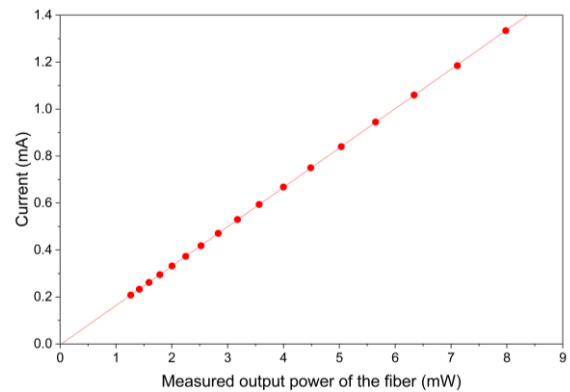


Fig. 6: Measured photocurrent of the InP PD as a function of the measured output power of the fiber, at -0.6 V reverse bias, leading to a fiber responsivity of 0.17 A/W .

Conclusion and outlook

We demonstrated the transfer of InP coupons using LIFT as novel integration method for PIC platform materials, with transfer accuracies approaching 500 nm (3σ) in both x- and y-directions. Besides showing the compatibility of this method with the transfer of thin and fragile InP coupons with a wide range of dimensions and aspect ratios, we demonstrated efficient electro-optical coupling between an InP PD and an SOI PIC integrated with LIFT.

These results show that LIFT has the potential to serve as the high-throughput, high-precision integration technology for PICs, enabling further mass adoption and industrialization of the PIC technologies that require hybrid or heterogeneous integration.

Acknowledgements

This project has received funding from the PhotonDelta National Growth Fund programme – www.photondelta.com. The work is carried out under the framework of Photonic Integration Technology Center (PITC). We would like to thank our colleagues Jacob Dalle, Sander Dorrestein, Ruben Pranger, Dorothee Hermes, Thijs Bel, Nils Bovée and Roy Verbeek for their help in the processing and Dr. James R. O’Callaghan (Tyndall National Institute) for the useful discussions on the InP coupons.

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